



Document Management System



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Title: Anaphylaxis

Definition: Anaphylaxis is a severe, acute andpotentially life-threatening medical condition in response to an allergenincluding: food which is the most common associated trigger, followed by(bee/wasp) stings and medications. When food is identified as the trigger,peanuts, tree nuts, fish, milk, eggs and shellfish (eg, shrimp, lobster, crab,scallops and oysters) are the products most often implicated in fatal ornear-fatal reactions.

Symptoms andsigns of anaphylaxis:

General/CNS	Irritability, drowsiness, lethargy, reduced level of consciousness, somnolence
Skin	Urticaria, pruritus, angioedema, flushing
Upper airway	Stridor, hoarseness, oropharyngeal or laryngeal edema, uvular edema, swollen lips/tongue, sneezing, rhinorrhea, upper airway obstruction
Lower airway	Coughing, dyspnea, bronchospasm, tachypnea, respiratory arrest
Cardiovascular	Tachycardia, hypotension, dizziness, syncope, arrhythmias, diaphoresis, pallor, cyanosis, cardiac arrest
Gastrointestinal	Nausea, vomiting, diarrhea, abdominal pain

Management:

Immediate action

2. Lay patient flat. Do not allow them to stand or walk. If breathing is difficult, allow them to sit.

3. Give INTRAMUSCULAR INJECTION (IMI) OF EPINEPHRINE (epinephrine) without delay using an epinephrine auto injector if available OR epinephrine ampoules and syringe.

- **Auto injector: 0.15 mg if weight < 30 kg or 0.3 mg if > 30 kg**
- **1:1000 IMI** into outer mid-thigh
- 0.01 mg per kg up to 0.5 mg per dose
- Repeat every 5 minutes as needed.
- If multiple doses required or a severe reaction consider epinephrine infusion

· ALWAYS give epinephrine FIRST, then asthma reliever if someone with known asthma and allergy to food, insects or medication has SUDDEN BREATHING DIFFICULTY (including wheeze, persistent cough or hoarse voice) even if there are no skin symptoms.

4. Monitor pulse, blood pressure, respiratory rate, pulse oximetry, conscious state.

5. Give high flow oxygen (6-8 L/min) and airway support if needed.

- Supplemental oxygen should be given to all patients with respiratory distress, reduced conscious level and those requiring repeated doses of epinephrine.
- Supplemental oxygen should be considered in patients who have asthma, other chronic respiratory disease, or cardiovascular disease.

6. If hypotensive:

During severe anaphylaxis with hypotension, marked fluid extravasation into the tissues can occur, do not forget fluid resuscitation

- Give intravenous normal saline (20 mL/kg rapidly under pressure), and repeat bolus if hypotension persists.
- Consider additional wide bore intravenous access.
- Monitor for signs of overtreatment (especially if respiratory distress or hypotension were absent initially) – including pulmonary oedema, hypertension.

Additional measures - IV epinephrine infusion

- **Caution - Intravenous boluses of epinephrine are NOT recommended due to risk of cardiac ischaemia or arrhythmia UNLESS the patient is in cardiac arrest.**
- **The protocol for adrenalin infusion:**
 - The dose is 0.1 µg/kg/min to 1 µg/kg/min (maximum 10 µg/min), with gradual titration of the infusion to produce a normal blood pressure
 - Monitor continuously – ECG and pulse oximetry and frequent non-invasive blood pressure measurements as a minimum to maximise benefit and minimise risk of overtreatment and epinephrine toxicity.

Additional measures to consider if IV epinephrine infusion is ineffective

For Upper airway obstruction

- Nebulized epinephrine (5 mL i.e. 5 ampoules of 1:1000).

For persistent hypotension/shock	· Give glucagon to patients with cardiogenic shock (specially who are taking beta-blockers) at a dose of 20 µg/kg to 30 µg/kg IV over 5 min (maximum dose 1 mg), followed by the initiation of a glucagon infusion at a rate of 5 µg/min to 15 µg/min, which is then titrated to effect. Beware of side effects including arrhythmias, severe hypotension and pulmonary edema. · Nor epinephrine infusion may be used in the critical care setting only with invasive blood pressure monitoring.
For persistent wheeze	Bronchodilators: · Salbutamol puffs of 100microgram using a spacer OR salbutamol by nebulizer. · Note: Bronchodilators do not prevent or relieve upper airway obstruction, hypotension or shock. Corticosteroids: · Oral prednisolone 1 mg/kg (maximum of 50 mg) or intravenous hydrocortisone 5mg/kg (maximum of 200 mg). · Note: Steroids must not be used as a first line medication in place of epinephrine.

Advanced airwaymanagement

- Oxygenation is more important thanintubation
- Always call for help from the mostexperienced person available
- If airway support is required, first usethe skills you are most familiar with (e.g. jaw thrust, Guedel or nasopharyngeal airway, bag-valve-mask with high flow oxygen attached). Thiswill save most patients, even those with apparent airway swelling (thesepatients have often stopped breathing due to circulatory collapse rather thanairway obstruction and can be adequately ventilated with basic life supportprocedures)
- DO NOT make prolonged attempts at intubation- remember the patient is not getting any oxygen while you are intubating.
- If unable to maintain an airway and thepatient's oxygen saturations are falling further approaches to the airway (e.g.cricothyrotomy) should be considered.

Special situation:Overwhelming anaphylaxis (cardiac arrest)

- Need both IV epinephrine bolus (cardiacarrest protocol) AND aggressive fluid resuscitation in addition to CPR (NormalSaline 20mL/kg stat, through a large bore IV under pressure, repeat if noresponse).
- Do not give up too soon - this is asituation when prolonged CPR should be considered, because the patient arrestedrapidly with previously normal tissue oxygenation, and has a potentiallyreversible cause.

Antihistamines

- Antihistamines have **no role intreating or preventing** respiratory or cardiovascular symptoms ofanaphylaxis

Post resuscitation care

- Observe for 4 h to 6 hafter the last dose of epinephrine.
- Some patients may needlonger observation eg 12 hrs including:
- Patients who live farfrom health facility.

- Who initially presented with more severe symptoms (eg, hypotension, severe respiratory distress)
- Patients who experience a biphasic reaction should be admitted to hospital for observation.
- Other patients who have high-risk features, such as peanut allergy, asthma or use of beta-blockers, should also be strongly considered for overnight observation or admission.
- Patients presenting with severe respiratory symptoms requiring definitive airway management or those who have persistent hypotension requiring IV epinephrine or glucagon infusions should be admitted to the intensive care unit.

Discharge management

- All patients suffering from anaphylaxis should be educated and counselled on avoidance of the precipitating trigger, and education about prevention of allergic reactions
- Referral to an allergist who will ensure that:
 - o Patients suffering from anaphylaxis should be given a self-injectable form of epinephrine.
 - o Parents, caretakers, older children and adolescents should be carefully educated about how to administer epinephrine
 - o Counselling about how to administer the drug when symptoms occur after exposure to the individual's known trigger.
 - o An epinephrine auto-injector should be kept with the child at all times.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
ASCA Guidelines - ADVANCED acute management of anaphylaxis.	ASCA	2015	
The management of anaphylaxis in childhood: position paper of The European academy of allergology and clinical immunology		2007	
Managing allergy, anaphylaxis: AAP releases customizable emergency plan	Julie Wang, M.D., FAAP	2017	

Written By: Dr. Khwater Abdelrahman

Checked By: Dr. Nashat Al Sukaiti

Authorized by: KHOULA JULENDA AL SAID

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